Computer Graphics

Projection Transformation

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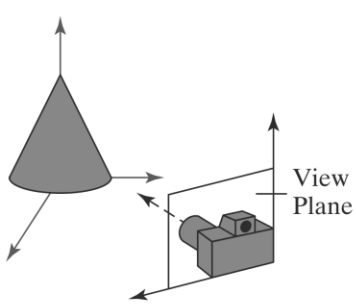
Viewing a Three-Dimensional Scene



- To obtain a display of a three-dimensional world-coordinate scene, we first set up a coordinate reference for the viewing, or "camera" parameters.
- This coordinate reference defines the position and orientation for a view plane (or projection plane, pp) that corresponds to a camera film plane .
- Object descriptions are then transferred to the viewing reference coordinates and projected onto the view plane.

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Projections: Projecting 3D World to 2D Image Plane

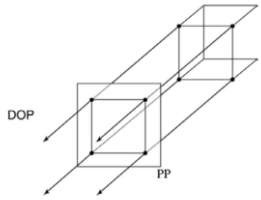
Projection involves projectors starting from 3D points and hitting the 2D projection plane, forming the image of the point

3

Types of Projections

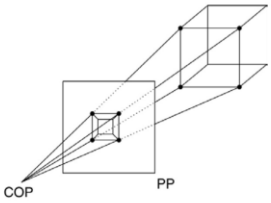
Parallel

- Projectors are parallel to each other, all have the same direction of projection (DOP).
- Project points on the object surface along parallel lines.



Perspective

- All projects pass through a point in space called the centre of projection (COP).
- Project points to the view plane along converging paths.



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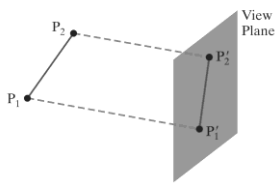
Types of Projections

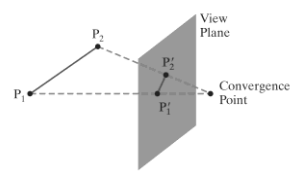
Parallel

- Coordinate positions are transferred to the view plane along parallel lines
- A parallel projection preserves relative proportions of objects, and this is the method used in computer aided drafting and design to produce scale drawings of three-dimensional objects



Perspective

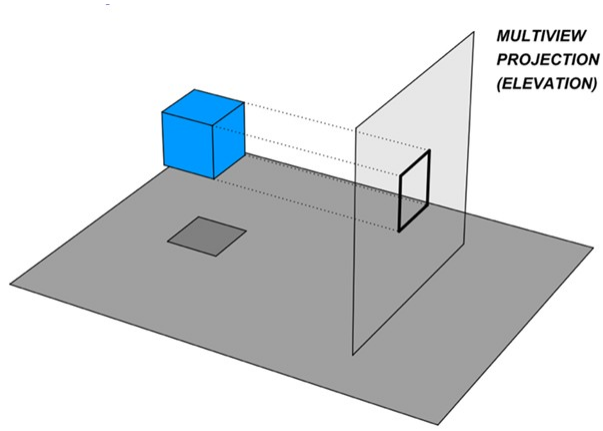
- Object positions are transformed to projection coordinates along lines that converge to a point behind the view plane
- Does not preserve relative proportions of objects. But perspective views of a scene are more realistic because distant objects in the projected display are reduced in size



Orthographic/Orthogonal Projection

Parallel Projection

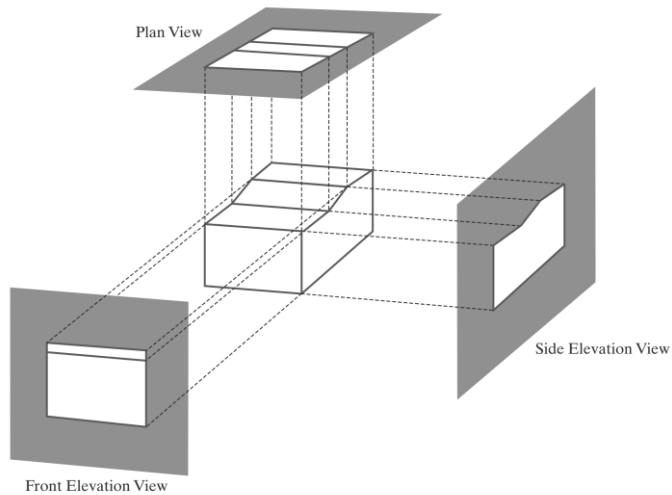
- A transformation of object descriptions to a view plane along lines that are all parallel to the view-plane normal vector
- Projection plane is perpendicular to the direction of projection.
- Lengths parallel to the projection plane are preserved.
- Only direction of projection matters; distance from the point to the projection plane doesn't.



Orthographic Projection

Parallel Projection

- Good approximation for a camera with a long focal length. (Orthographic with uniform scaling).
- If direction of projection parallel to the axes: plan, elevation, side elevation.



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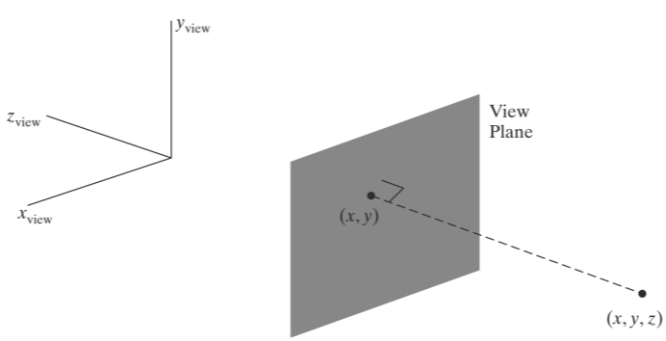
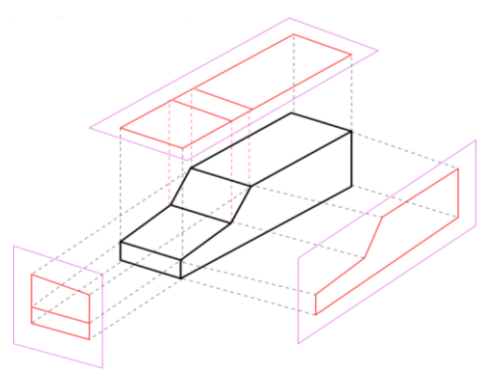
Orthographic Projection

Parallel Projection



$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$



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Axonometric Orthogonal Projections

Parallel Projection

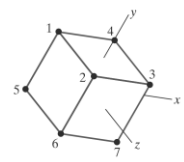
Projections that display more than one face of an object is Axonometric

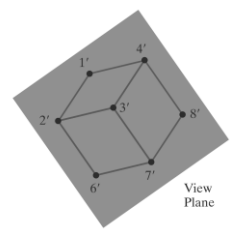
Isometric projection

The most commonly used axonometric projection is the isometric projection, which is generated by aligning the projection plane (or the object) so that the plane intersects each coordinate axis (in which the object is defined, called the principal axes) at the same distance from the origin

Cube Example

- Obtained by aligning the view plane normal vector along a cube diagonal.
- There are eight positions, one in each **octant**, for obtaining an isometric view. All three principal axes are foreshortened equally in an isometric projection, so that relative proportions are maintained





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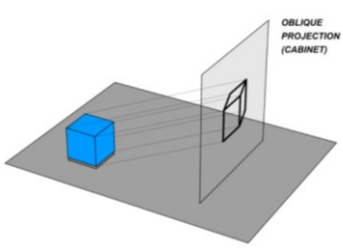
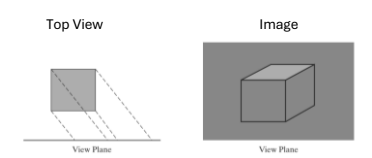
Oblique Parallel Projection

Parallel Projection





- A parallel-projection view of a scene is obtained by transferring object descriptions to the view plane along projection paths that can be in **any selected direction** relative to the view-plane normal vector.
- When the projection path is not perpendicular to the view plane
- Using this projection, we can produce combinations such as a front, side, and top view of an object.
- Oblique parallel projections are defined by a vector direction for the projection lines, and this direction can be specified in various ways.

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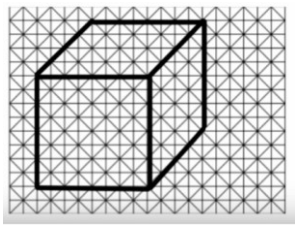
Oblique Projection

Parallel Projection

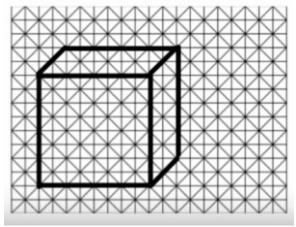
Cavalier

- Projectors make 45 degrees with the projection plane
- Length along the depth axis is preserved



Cabinet

- Projectors make $\arctan(2)$ degrees with the projection plane
- Length along depth axis is halved



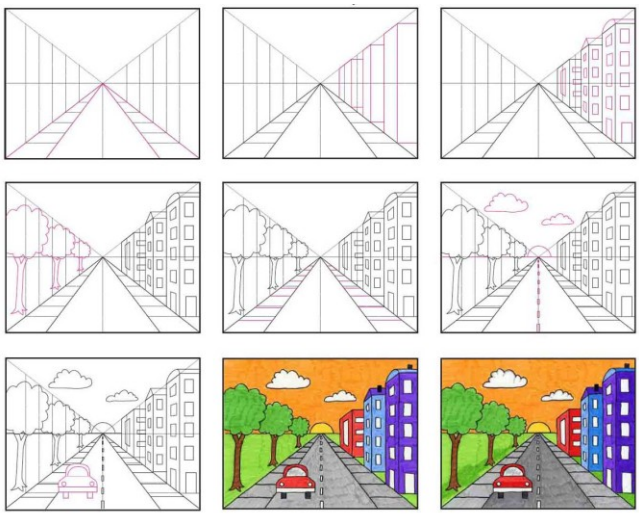
More Realistic

Perspective Projection

The number of vanishing points depends on how many of the object's principal directions are not parallel to the projection plane

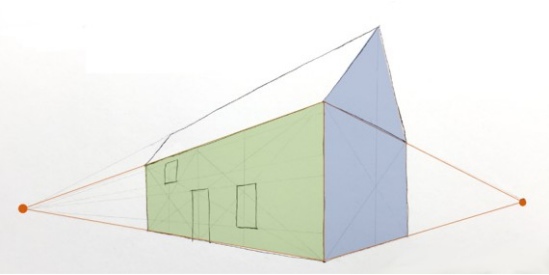
1-point Perspective

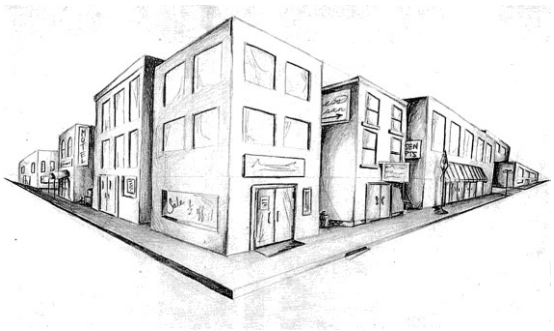
- Two principal directions are parallel to the image plane, so only the third direction "converges" to one vanishing point



2-point Perspective

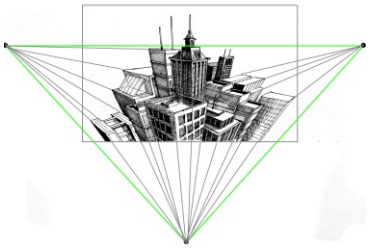
- One principal direction is parallel to the image plane, so the other two converge to two vanishing points

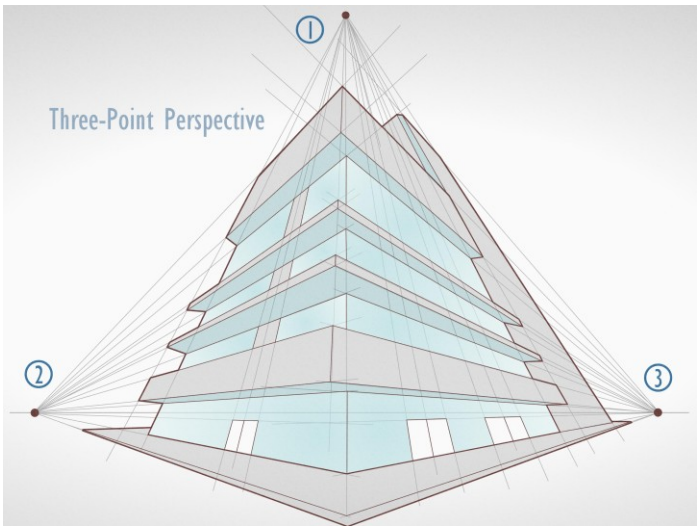




3-point Perspective

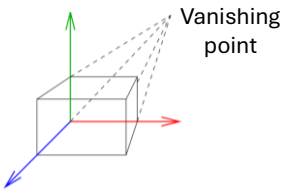
- None of the principal directions are parallel to the image plane, so all three converge



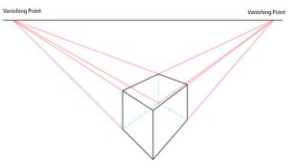


Projection Geometry

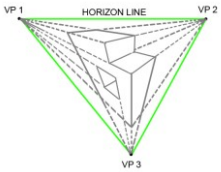
1-point Perspective



2-point Perspective



3-point Perspective

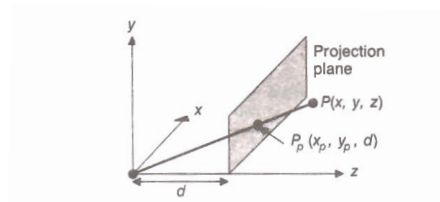


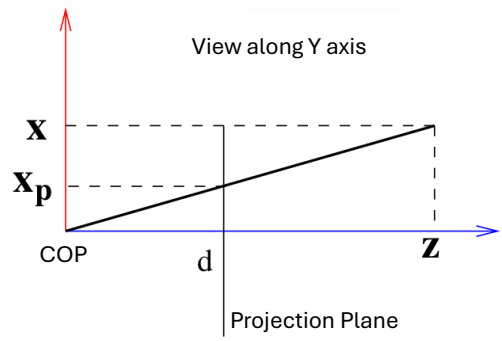
Geometry of Perspective Projection

- $\frac{x_p}{d} = \frac{x}{z}$, $\frac{y_p}{d} = \frac{y}{z}$, $z = d$
- In matrix form

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/d & 0 \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

- Coordinates scaled down proportional to the depth or z values.



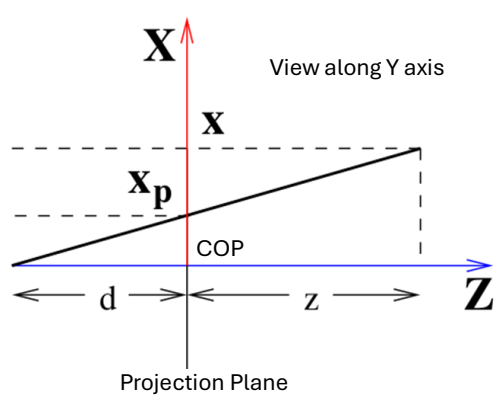


Geometry of Perspective Projection

- Shift origin to lie on the projection plane, COP at $(0,0,-d)$
- $x_p = \frac{Xd}{Z+d}$, $y_p = \frac{Yd}{Z+d}$, $z_p = 0$
- We get the matrix:

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1/d & 1 \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

- Orthographic Project matrix is a special case when $d \to \infty$



Projections Summary

Perspective

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1/d & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$d < \infty$$

Orthographic

$$\begin{bmatrix} x_p \\ y_p \\ z_p \\ w_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$d \to \infty$$

Volume of Visibility

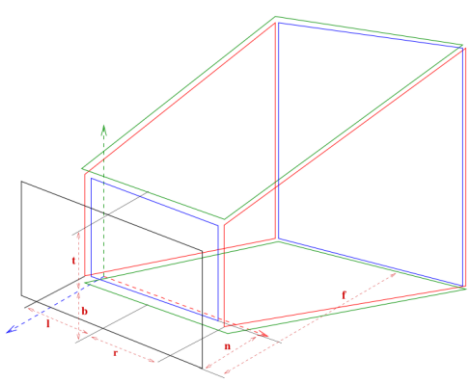
- Cameras have finite fields of view in horizontal and vertical directions.
- What is the shape of its visible space?
- A cylinder for orthographic projections and a cone starting from the COP for perspective may seem natural.
- Mathematics is difficult for cones; rectangular structures are easier!
- **View Volume**: The volume of potentially visible space.

View Volume

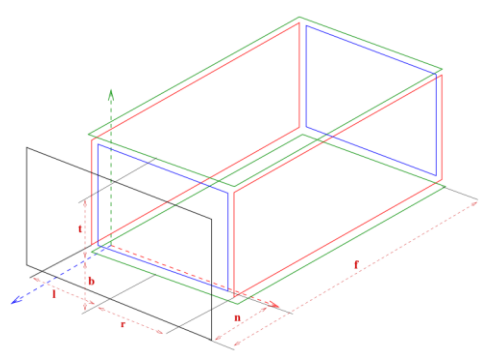
- View volume is a cube for orthographic cameras and a (truncated) pyramid for perspective projections.
- 4 planes (left, right, top, bottom) define the view volume.
- Graphics cameras use 2 additional planes to limit visibility: near & far!
- Planes are specified in VRC; they move with the camera

Projections Summary

Perspective

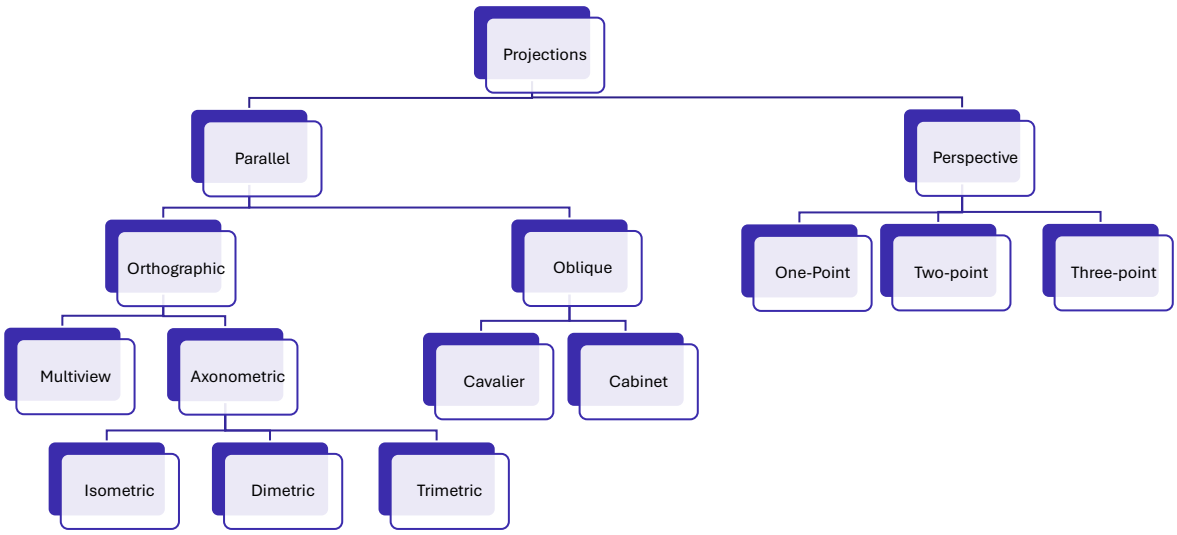


Orthographic



View volume: Top, Bottom, Left, Right, Near, Far

Taxonomy of different Projection Types



Thank You

Next: Viewport Transformations